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B. Tech. Degree V Semester Examination November 2014**CS 1505 DATABASE MANAGEMENT SYSTEMS**
(2012 Scheme)

Time: 3 Hours

Maximum Marks: 100

PART A(Answer *ALL* questions)

(8 x 5 = 40)

- I. (a) Differentiate between total participation constraint and partial participation constraint, with the help of an example.
- (b) Describe, with the help of an example, how can we represent ternary relationship with the E-R diagram.
- (c) Give any two advantages of a sparse index over a dense index.
- (d) Why is the B+ Tree preferable over the B Tree?
- (e) What do you mean by a self referencing foreign key? Explain with an example.
- (f) Define the term data integrity. Explain various types.
- (g) Discuss the two different forms of schedule equivalence.
- (h) What is the difference between lock based techniques and time stamp based techniques?

PART B

(4 x 15 = 60)

- II. (a) Define the term relationship. Illustrate the difference between relationship type and relationship instance. (8)
- (b) What are the advantages and disadvantages of a DBMS? (7)
- OR**
- III. (a) Explain the various types of attributes. What are the various situations in which an attribute can use a null value? (8)
- (b) Explain the characteristics of various record based data models. (7)
- IV. (a) Discuss the commonly used hash functions. What is the main problem associated with most of them, and how can it be resolved? (8)
- (b) What is multi-level index? Also give reasons for having multilevel index. (7)
- OR**
- V. (a) What does the order of a B-Tree and B+ - Tree indicate? Also explain the structures of both internal and external nodes of a B-tree and B+ tree. (8)
- (b) Explain two techniques for implementing a heap file, with the help of diagrams. (7)

(P.T.O.)

- VI. (a) What is the outer join? Explain how does it differ from the inner join. (7)
 (b) Express the following query in tuple relational calculus. (8)
 (i) Retrieve the Name, Address and Phoneno of all publishers located in New York State.
 (ii) Retrieve the title and price of all text books with a page count greater than 600.

OR

- VII. (a) Consider the universal relation R (A,B,C,D,E,F,G,H,I,J) and the set of functional dependencies. (7)
 $AB \rightarrow C$
 $A \rightarrow DE$
 $B \rightarrow F$
 $F \rightarrow GH$
 $D \rightarrow IJ$
 What is the key for R? Decompose R into 2NF and 3NF relations.
 (b) Express the following query in domain relational calculus. (8)
 (i) Retrieve the Name, Address and phoneno of all publishers located in New York State.
 (ii) Retrieve the title and price of all text books with a page count greater than 600.

- VIII. (a) Discuss basic time stamp ordering. How is it different from strict time stamp ordering protocol? (8)
 (b) How is a transaction validated in an optimistic concurrency control technique? (7)

OR

- IX. (a) Describe "wait-die" and "wound-wait" techniques of deadlock prevention. (8)
 (b) What is data mining? Why are data in data warehouses more suitable for the data mining process? (7)
